

1. We apply EROs as follows.

$$\begin{aligned}
 A = \begin{pmatrix} 2 & 3 & a & b \\ 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix} &\xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & a & b \\ 4 & 3 & 2 & 1 \end{pmatrix} \xrightarrow{\substack{-2R_1+R_2 \rightarrow R_2 \\ -4R_1+R_3 \rightarrow R_3}} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & a-6 & b-8 \\ 0 & -5 & -10 & -15 \end{pmatrix} \\
 &\xrightarrow{-\frac{1}{5}R_3 \rightarrow R_3} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -1 & a-6 & b-8 \\ 0 & 1 & 2 & 3 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & -1 & a-6 & b-8 \end{pmatrix} \\
 &\xrightarrow{\substack{-2R_2+R_1 \rightarrow R_1 \\ R_2+R_3 \rightarrow R_3}} \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & a-4 & b-5 \end{pmatrix}.
 \end{aligned}$$

Clearly, there are at least two leading ones. So $\text{rank } A$ can only be 2 or 3. Correspondingly, $\dim N(A)$ can only be $4 - 2 = 2$ or $4 - 3 = 1$.

Both are possible. For example, when $(a, b) = (4, 5)$, $\text{rank}(A) = 2$ and $\dim(N(A)) = 2$. When $(a, b) = (5, 5)$, $\text{rank}(A) = 3$ and $\dim(N(A)) = 1$.

2. (a)

$$\begin{aligned}
 V &= \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} : x_2 = x_4 \text{ and } x_1, x_2, x_3, x_4 \in \mathbb{R} \right\} \\
 &= \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_2 \end{pmatrix} : x_1, x_2, x_3 \in \mathbb{R} \right\} \\
 &= \left\{ x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} : x_1, x_2, x_3 \in \mathbb{R} \right\} \\
 &= \{x_1 \mathbf{e}_1 + x_2(\mathbf{e}_2 + \mathbf{e}_4) + x_3 \mathbf{e}_3 : x_1, x_2, x_3 \in \mathbb{R}\} \\
 &= \text{Span}(\{\mathbf{e}_1, \mathbf{e}_3, \mathbf{e}_2 + \mathbf{e}_4\})
 \end{aligned}$$

is a subspace of $\mathbb{R}^4$.

(b) The set $S = \{\mathbf{e}_1, \mathbf{e}_2 + \mathbf{e}_4, \mathbf{e}_2 + \mathbf{e}_3 + \mathbf{e}_4\}$ is a basis for V because

(i) S is a linearly independent set: suppose that

$$c_1 \mathbf{e}_1 + c_2(\mathbf{e}_2 + \mathbf{e}_4) + c_3(\mathbf{e}_2 + \mathbf{e}_3 + \mathbf{e}_4) = \mathbf{0}_{4 \times 1},$$

that is,

$$(c_1, c_2 + c_3, c_3, c_2 + c_3)^T = (0, 0, 0, 0)^T.$$

Then $c_1 = c_2 + c_3 = c_3 = 0$ and hence $c_1 = c_2 = c_3 = 0$.

(ii)

$$\begin{aligned} V &= \{x_1 \mathbf{e}_1 + x_2(\mathbf{e}_2 + \mathbf{e}_4) + x_3 \mathbf{e}_3 : x_1, x_2, x_3 \in \mathbb{R}\} \quad (\text{by (a)}) \\ &= \{x_1 \mathbf{e}_1 + (x_2 - x_3)(\mathbf{e}_2 + \mathbf{e}_4) + x_3(\mathbf{e}_2 + \mathbf{e}_3 + \mathbf{e}_4) : x_1, x_2, x_3 \in \mathbb{R}\} \\ &= \{x_1 \mathbf{e}_1 + y(\mathbf{e}_2 + \mathbf{e}_4) + x_3(\mathbf{e}_2 + \mathbf{e}_3 + \mathbf{e}_4) : x_1, y, x_3 \in \mathbb{R}\} \\ &= \text{span}(S) \end{aligned}$$

- (c) Let a_1, a_2, a_3, a_4 be the columns of A . Suppose that $V = N(A)$. Then $N(A)$ contains the vectors $\mathbf{e}_1, \mathbf{e}_2 + \mathbf{e}_4, \mathbf{e}_3 \in V$, we must have $A\mathbf{e}_1 = A(\mathbf{e}_2 + \mathbf{e}_4) = A\mathbf{e}_3 = \mathbf{0}_{n \times 1}$. This implies $a_1 = a_2 + a_4 = a_3 = \mathbf{0}_{n \times 1}$. In other words,

$$A = \begin{pmatrix} 0_{n \times 1} & a & 0_{n \times 1} & -a \end{pmatrix}$$

for some $a \in \mathbb{R}^n$.

Conversely, suppose that $A = \begin{pmatrix} 0_{n \times 1} & a & 0_{n \times 1} & -a \end{pmatrix}$ for some $a \in \mathbb{R}^n$.

If $a = \mathbf{0}_{n \times 1}$, it is clear that the null space of $A = \mathbf{0}_{n \times 4}$ is $\mathbb{R}^4$, which is not V .

If $a \neq \mathbf{0}_{n \times 1}$, we see that $Ax = \mathbf{0}_{n \times 1}$ if and only if $x_2 a - x_4 a = \mathbf{0}_{n \times 1}$ (where $x = (x_1, x_2, x_3, x_4)^T$), which holds if and only if $x_2 = x_4$. This implies $N(A) = V$.

Therefore, A can be any matrix of the form $\begin{pmatrix} 0_{n \times 1} & a & 0_{n \times 1} & -a \end{pmatrix}$ with $a \neq \mathbf{0}_{n \times 1}$.

3. (a) Note that $\text{col}(BA) \subset \text{col}(B)$ for any $n \times n$ matrix B .

($\supset$): $\text{col}(A^2) \subset \text{col}(A)$, by taking $B = A$.

($\subset$): $\text{col}(A) = \text{col}(A^3) \subset \text{col}(A^2)$, by taking $B = A^2$.

- (b) Note that $A^2 = A^4 = \dots = A^{2022} = A^{2024}$. So

$$\begin{aligned} \text{rank}(A^{2024}) &= \text{rank}(A^2) \\ &= \dim(\text{col}(A^2)) \\ &= \dim(\text{col}(A)) \quad (\text{by (a)}) \\ &= \text{rank}(A). \end{aligned}$$

- (c) For any $y \in \mathbb{R}^n$,

$$\begin{aligned} y \in \text{col}(A^2 + A) &\implies y = (A^2 + A)x \text{ for some } x \in \mathbb{R}^n \\ &\implies (A - I_n)y = (A - I_n)(A^2 + A)x = (A^3 - A)x = \mathbf{0} \\ &\implies y \in N(A - I_n). \end{aligned}$$

Thus $\text{col}(A^2 + A) \subset N(A - I_n)$.

4. (a) Proof: For any $n \times n$ matrix A , $\text{row}(BA) = \text{row}(A)$ if B is invertible. So

$$\text{rank}(BA) = \dim(\text{row}(BA)) = \dim(\text{row}(A)) = \text{rank}(A).$$

- (b) Proof: Take $A = I_n$, we have $\text{rank}(B) = \text{rank}(I_n) = n$ and hence the reduced row echelon form of B is I_n . Thus B is invertible.

- (c) Disproof: Take $A = B = \mathbf{0}_{n \times n}$, then $\text{rank}(BA) = 0 = \text{rank}(A)$, and $B = \mathbf{0}_{n \times n}$ is not invertible.